

Participation Activity 2

Last Name: _____

First Name: _____

Std #: _____

The function prototype of **func** is: `int func(int x, int y);`

The body of **func** is:

```
func:
    movl %edi, %eax
    subl %esi, %eax
    ret
```

In our C program, we want to call **func** with the arguments **3** and **-2** such that **func** returns **-5** as its returned value. In which order must the arguments **3** and **-2** be placed when we call **func**, i.e., which value must **x** contain? which value must **y** contain?

Your answer: `int result = func(_____, _____);`

Participation Activity 2 - SOLUTION

Answer: `int result = func(-2,3);`

x must contain -> **-2**

y must contain -> **3**

x in edi, y in esi, result (x - y) in eax

func:

```
    movl %edi, %eax    # eax <- x
    subl %esi, %eax    # eax <- eax(x) - y
    ret
```

If x = -2 and y = 3 => -2 - 3 => -5

If x = 3 and y = -2 => 3 - (-2) => 5